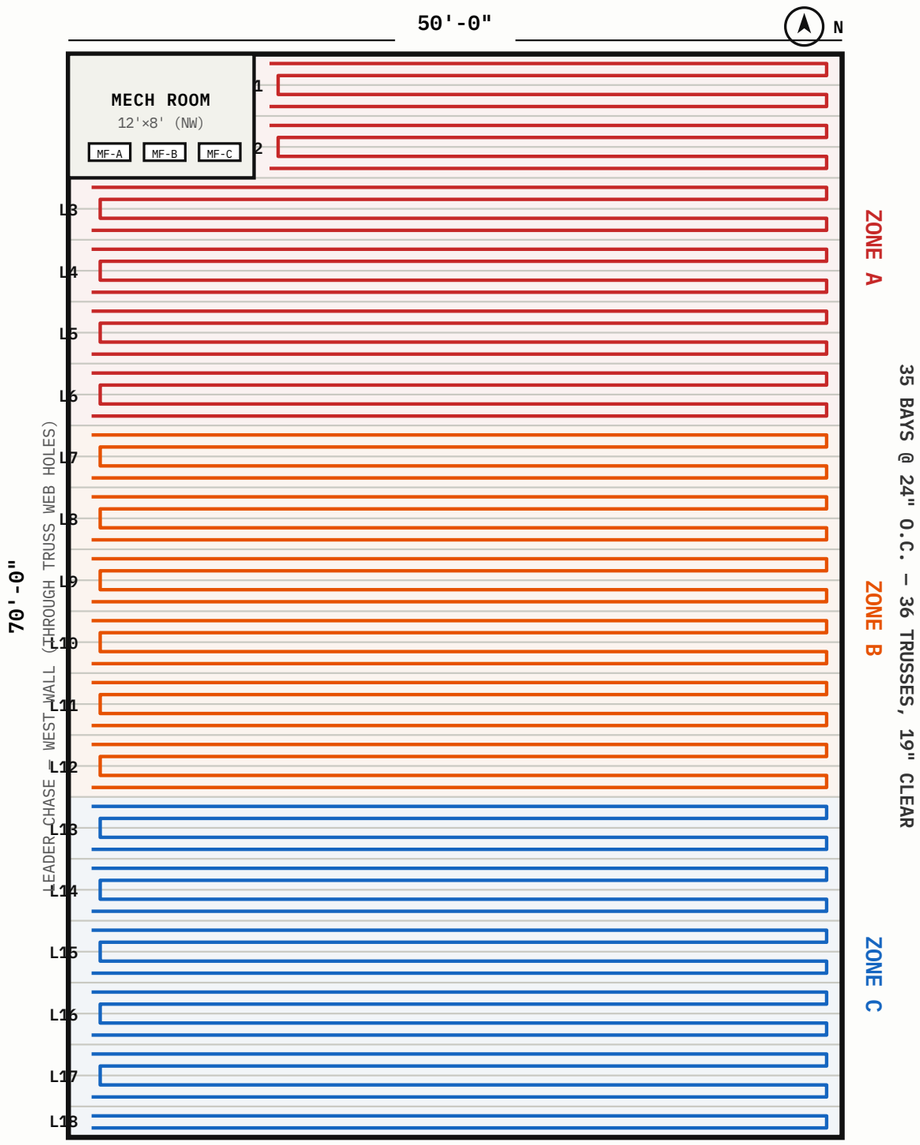
IN-FLOOR RADIANT LAYOUT — 50' × 70' SHOP

1/2" O2-BARRIER PEX, DOWN-AND-BACK EVERY TRUSS BAY • 2 BAYS PER LOOP • NOT TO SCALE



LOOP SCHEDULE

LOOP	ZONE	BAYS	ACTIVE FT	LEADER FT ×2	TOTAL FT
L1	A	1–2	152	6	**164**
L2	A	3–4	152	10	**172**
L3	A	5–6	200	14	**228**
L4	A	7–8	200	18	**236**
L5	A	9–10	200	22	**244**
L6	A	11–12	200	26	**252**
L7	B	13–14	200	30	**260**
L8	B	15–16	200	34	**268**
L9	B	17–18	200	38	**276**
L10	B	19–20	200	42	**284**
L11	B	21–22	200	46	**292**
L12	B	23–24	200	50	**300**
L13	C	25–26	200	54	**308 †**
L14	C	27–28	200	58	**316 †**
L15	C	29–30	200	62	**324 †**
L16	C	31–32	200	66	**332 †**
L17	C	33–34	200	70	**340 †**
L18	C	35 single	100	74	**248**

† OVER 300' — RUN 3/4" PEX LEADERS FROM MANIFOLD TO FIRST BAY, OR SET FLOW METER TO 0.4 GPM. L1–L2 SHORTENED BY MECH ROOM FOOTPRINT (12'×8').

BAY MATH

TRUSSES SPAN 50' SIDE, 24" O.C. ALONG 70' WALL
70' = 840" ÷ 24" = 35 BAYS (36 TRUSSES)
CLEAR BAY WIDTH: 24" − 5" WEB = 19"
2 PASSES/BAY ≈ 9.5" TUBE SPACING
ZONES: A = BAYS 1–12, B = 13–24, C = 25–35 (ODD → L18 SINGLE-BAY)

ZONE SUMMARY

ZONE	BAYS	LOOPS	PEX FT
A	1–12	L1–L6	1296
B	13–24	L7–L12	1680
C	25–35	L13–L18	1868
TOTAL (18 LOOPS @ 0.5 GPM = 9 GPM)			**4844**

DESIGN NOTES

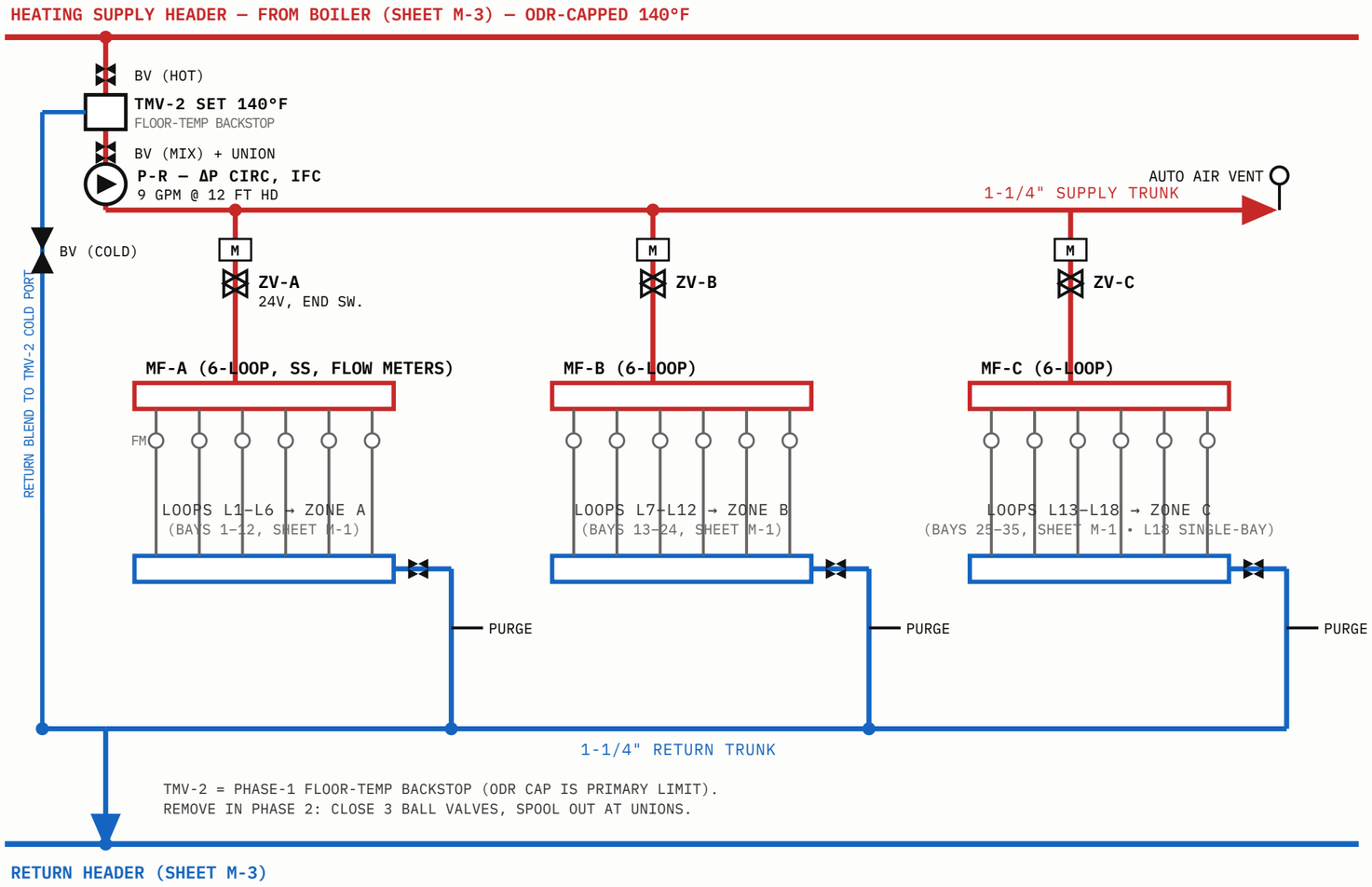
1. SUSPEND PEX WITH EXTRUDED ALUMINUM HEAT-TRANSFER PLATES, TUBE 2" BELOW SUBFLOOR. R-19 MIN BATT BELOW TUBE IN EACH BAY.
2. DRILL TRUSS WEBS PER TRUSS MFR CHART ONLY — NO CHORD PENETRATIONS. LEADERS CROSS BAYS THROUGH WEB HOLES ALONG WEST WALL CHASE.
3. NO FITTINGS OR SPLICES BELOW FLOOR. ONE CONTINUOUS COIL PER LOOP, HOME-RUN TO ZONE MANIFOLD (SHEET M-2).
4. SLEEVE PEX AT EVERY WEB PENETRATION; PROTECT PLATES 18" FROM ANY FLUE/LIGHT FIXTURE HEAT SOURCE.
5. PRESSURE TEST 60 PSI AIR, 24 HR, BEFORE INSULATION AND CEILING CLOSE-IN.
6. MAX SUPPLY WATER TEMP 140°F (BOILER ODR CAP, SHEET M-3/M-4). FLOOR SURFACE NOT TO EXCEED 85°F.

BILL OF MATERIALS — SHEET M-1

1/2" O2-BARRIER PEX-B, 500' COILS	11 COILS (4844' + 5% WASTE)
ALUM. HEAT-TRANSFER PLATES, 4'	≈ 860 EA (80% COVERAGE)
R-19 UNFACED BATT, 24" W	≈ 3,300 SF
PEX SLEEVES / GROMMETS	≈ 150 EA
3/4" PEX (ZONE C LONG LEADERS)	≈ 700'

MECHANICAL ROOM — IN-FLOOR PIPING

ONE PUMP + THREE ZONE VALVES • ONE MANIFOLD PER ZONE • CROSSINGS NOT CONNECTED W/O DOT • NTS



PUMPING: COST COMPARISON

	A: 1 PUMP + 3 ZV ★	B: 3 PUMPS
PUMPS	1× ΔP CIRC ≈ $230	3× CIRC ≈ $450
VALVES	3× ZV ≈ $285	—
FLANGES/CHECKS	1 SET ≈ $40	3 SETS ≈ $120
RELAY/WIRING	≈ $50	≈ $110
TOTAL	**≈ $605**	**≈ $680**
RUN POWER	5–45 W	15–130 W

★ DRAWN. ΔP-ADAPTIVE PUMP SELF-TRIMS AS ZONE VALVES OPEN/CLOSE — NO BYPASS NEEDED.

CONTROL LOGIC

1. ZONE T-STAT CALL → OPEN ITS ZV (24VAC).
2. ZV END SWITCH (ANY) → ENABLE P-R. PUMP RUNS ON ΔP AUTO.
3. ALL END SWITCHES OPEN → P-R OFF.
4. PHASE 2 ADDS BURST/COOLING LOCKOUT OF RADIANT (SHEET M-4).
5. SUPPLY LIMITED TO 140°F BY NAVIEN OUTDOOR-RESET CAP; TMV-2 AT TRUNK IS THE MECHANICAL BACKSTOP (SET 140°F).

NOTES

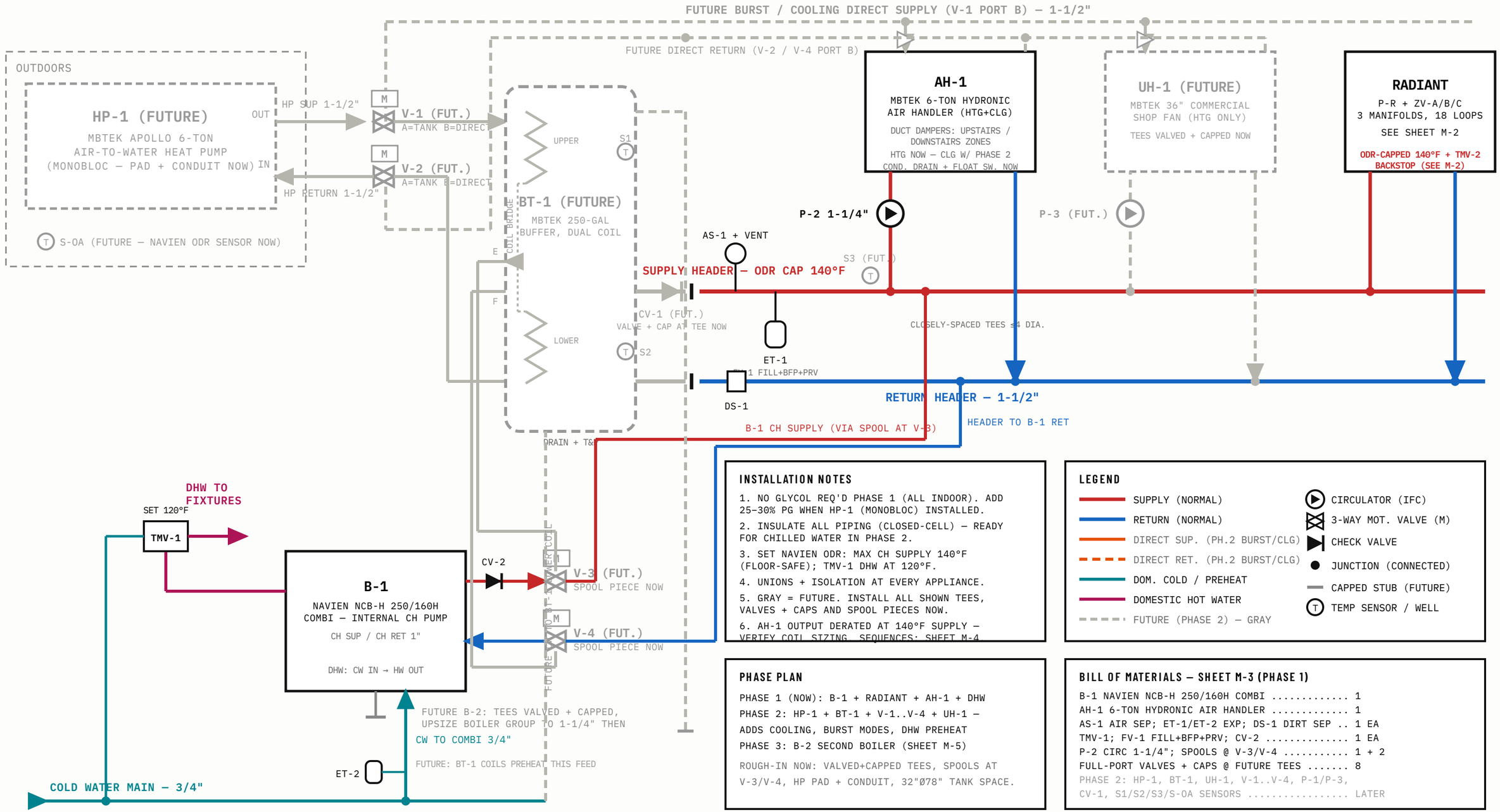
1. MANIFOLDS WALL-MOUNTED IN MECH ROOM (NW CORNER), LEVEL, SUPPLY BAR UP. LABEL EVERY LOOP (L1–L18).
2. FLOW METERS SET 0.5 GPM/LOOP (0.4 GPM WHERE NOTED, SHEET M-1).
3. MANUAL AIR VENT + FILL/DRAIN CAP EACH MANIFOLD END; AUTO VENT AT TRUNK HIGH POINT.
4. ISOLATION VALVES BOTH SIDES OF PUMP AND EACH MANIFOLD; PURGE COCK EACH RETURN.
5. THERMOMETER WELLS ON EACH MANIFOLD SUPPLY + RETURN.

BILL OF MATERIALS — SHEET M-2

6-LOOP SS MANIFOLD W/ FLOW METERS	3
ΔP CIRCULATOR W/ IFC (P-R)	1
24V ZONE VALVE W/ END SWITCH, 1"	3
1-1/4" TRUNK PIPE + FITTINGS	≈ 40'
FULL-PORT BALL VALVES 1"–1-1/4"	10
PURGE COCKS / BOILER DRAINS	3
TMV-2 1-1/4" THERMOSTATIC MIX VALVE + 3 BV + UNIONS	1 SET
AUTO AIR VENT + 2× DIAL THERMOMETERS/MFLD	1 + 6

MECHANICAL ROOM — MAJOR PIPING

PHASE 1: COMBI + RADIANT + AH-1 • GRAY = FUTURE (HP-1, BT-1, UH-1) — PLUMB TEES VALVED + CAPPED NOW • NTS



INSTALLATION NOTES

1. NO GLYCOL REQ'D PHASE 1 (ALL INDOOR). ADD 25–30% PG WHEN HP-1 (MONOBLOC) INSTALLED.
2. INSULATE ALL PIPING (CLOSED-CELL) — READY FOR CHILLED WATER IN PHASE 2.
3. SET NAVIEN ODR: MAX CH SUPPLY 140°F (FLOOR-SAFE); TMV-1 DHW AT 120°F.
4. UNIONS + ISOLATION AT EVERY APPLIANCE.
5. GRAY = FUTURE. INSTALL ALL SHOWN TEES, VALVES + CAPS AND SPOOL PIECES NOW.
6. AH-1 OUTPUT DERATED AT 140°F SUPPLY — VERIFY COIL SIZING. SEQUENCES: SHEET M-4.

LEGEND

- SUPPLY (NORMAL)
- RETURN (NORMAL)
- DIRECT SUP. (PH.2 BURST/CLG)
- DIRECT RET. (PH.2 BURST/CLG)
- DOM. COLD / PREHEAT
- DOMESTIC HOT WATER
- FUTURE (PHASE 2) — GRAY
- CIRCULATOR (IFC)
- 3-WAY MOT. VALVE (M)
- CHECK VALVE
- JUNCTION (CONNECTED)
- CAPPED STUB (FUTURE)
- TEMP SENSOR / WELL

PHASE PLAN

PHASE 1 (NOW): B-1 + RADIANT + AH-1 + DHW

PHASE 2: HP-1 + BT-1 + V-1..V-4 + UH-1 — ADDS COOLING, BURST MODES, DHW PREHEAT

PHASE 3: B-2 SECOND BOILER (SHEET M-5)

ROUGH-IN NOW: VALVED+CAPPED TEES, SPOOLS AT V-3/V-4, HP PAD + CONDUIT, 32"Ø78" TANK SPACE.

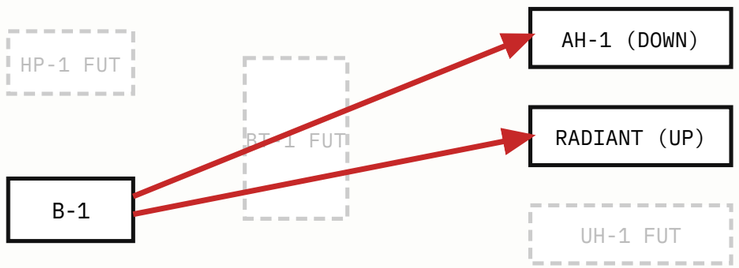
BILL OF MATERIALS — SHEET M-3 (PHASE 1)

B-1 NAVIEN NCB-H 250/160H COMBI	1
AH-1 6-TON HYDRONIC AIR HANDLER	1
AS-1 AIR SEP; ET-1/ET-2 EXP; DS-1 DIRT SEP	1 EA
TMV-1; FV-1 FILL+BFP+PRV; CV-2	1 EA
P-2 CIRC 1-1/4"; SPOOLS @ V-3/V-4	1 + 2
FULL-PORT VALVES + CAPS @ FUTURE TEES	8
PHASE 2: HP-1, BT-1, UH-1, V-1..V-4, P-1/P-3, CV-1, S1/S2/S3/S-OA SENSORS	LATER

HEATING & COOLING STRATEGIES — SEQUENCE OF OPERATION

PHASE 1 (NOW): NAVIEN ODR + ZVC404 • PHASE 2 (FUTURE): HP-1 + BT-1 + HBX • DHW VIA COMBI, ALWAYS

HEATING — PHASE 1 (NOW)

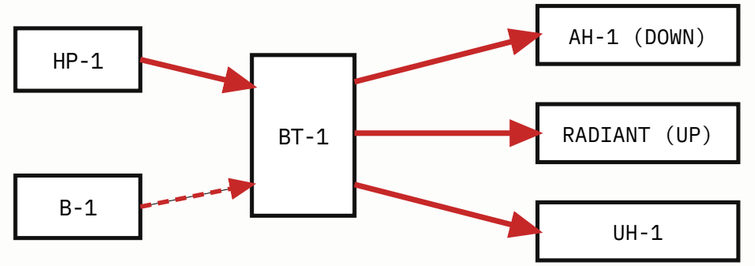


- B-1 SOLE HEAT SOURCE. NAVIEN INTERNAL OUTDOOR RESET (SUPPLIED SENSOR), CH MAX 140°F — FLOOR-SAFE, TMV-2 BACKSTOP (M-2).
- RADIANT HEATS UPSTAIRS (T1–T3 → ZV-A/B/C); AH-1 HEATS DOWNSTAIRS (T4 + D-UP/D-DN DAMPERS).
- ANY ZV END SWITCH OR T4 CALL → B-1 TT + PUMP ENABLE.
- CLOSELY-SPACED TEES DECOUPLE B-1 INTERNAL PUMP FROM P-R / P-2.

DHW & COOLING — PHASE 1 (NOW)

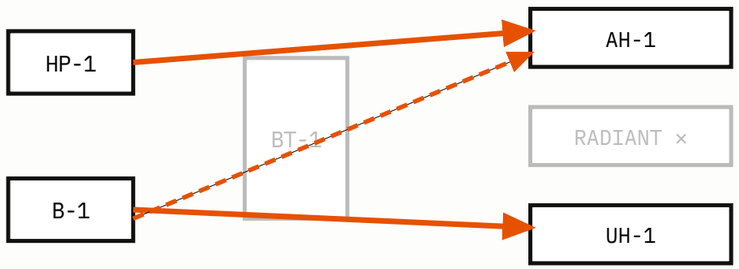
- DHW: COMBI INTERNAL PRIORITY. CW MAIN → B-1 → TMV-1 SET 120°F → FIXTURES. ET-2 ON COLD FEED.
- FUTURE: BT-1 DUAL COILS PREHEAT THE CW FEED (GRAY ROUTE, SHEET M-3) — CUTS DHW GAS USE.
- COOLING: NONE IN PHASE 1 — NO CHILLED SOURCE UNTIL HP-1. AH-1 RUNS HEAT-ONLY.
- ROUGHED IN NOW FOR COOLING: AH CONDENSATE DRAIN + FS-1 FLOAT SWITCH, CLOSED-CELL INSULATION ON ALL HYDRONIC PIPING, VALVED + CAPPED DIRECT-LINE TEES.
- AH-1 COIL DERATED AT 140°F SUPPLY (BELOW ITS 180°F RATING) — VERIFY HEAT LOSS; HOTTER DIRECT SUPPLY ARRIVES WITH PHASE 2 BURST PIPING.

FUTURE — NORMAL OPS W/ HP-1 + BT-1



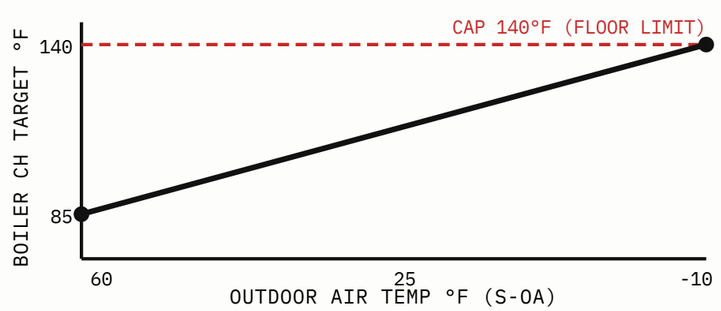
- HP-1 CHARGES (OR CHILLS) BT-1 ON OUTDOOR RESET VIA HBX (S1/S2); TANK CAPPED 120°F HTG / 45°F CLG.
- TANK MASS ENDS SHORT-CYCLING; B-1 DROPS TO 2ND STAGE ONLY.
- COOLING: AH-1 DRAWS CHILLED WATER; RADIANT + UH-1 LOCKED OUT (SWEAT).
- DHW PREHEAT VIA TANK COILS; SOLAR: HA FORCES HP-1 TO CHARGE TANK ON PV EXCESS.

FUTURE — BURST HTG & CLG



- TRIGGER: ROOM ≥ 4°F FROM SETPOINT ON A CALL.
- HTG: ALL VALVES → B; HP-1 + B-1 MAX-TEMP WATER DIRECT TO AH-1/UH-1, TANK BYPASSED.
- CLG: V-1/V-2 → B; COLDEST HP WATER DIRECT TO AH-1.
- RADIANT ZVs FORCED CLOSED — DIRECT TEMPS EXCEED FLOOR LIMIT.
- EXIT: WITHIN 1°F OR 45-MIN TIMEOUT → VALVES SPRING TO A.

OUTDOOR RESET & SETPOINTS



CH SUPPLY MAX (NAVIEN ODR CAP)	140°F
RESET: 60°F OAT → 85°F / -10°F OAT → CAP	LINEAR
DHW AT TMV-1	120°F
FUTURE BURST TRIGGER (PH.2)	Δ 4°F
FUTURE BURST EXIT / TIMEOUT	Δ 1°F / 45 MIN
FUTURE CHILLED TANK TARGET	45°F

CONTROLS, WIRING & FUTURE

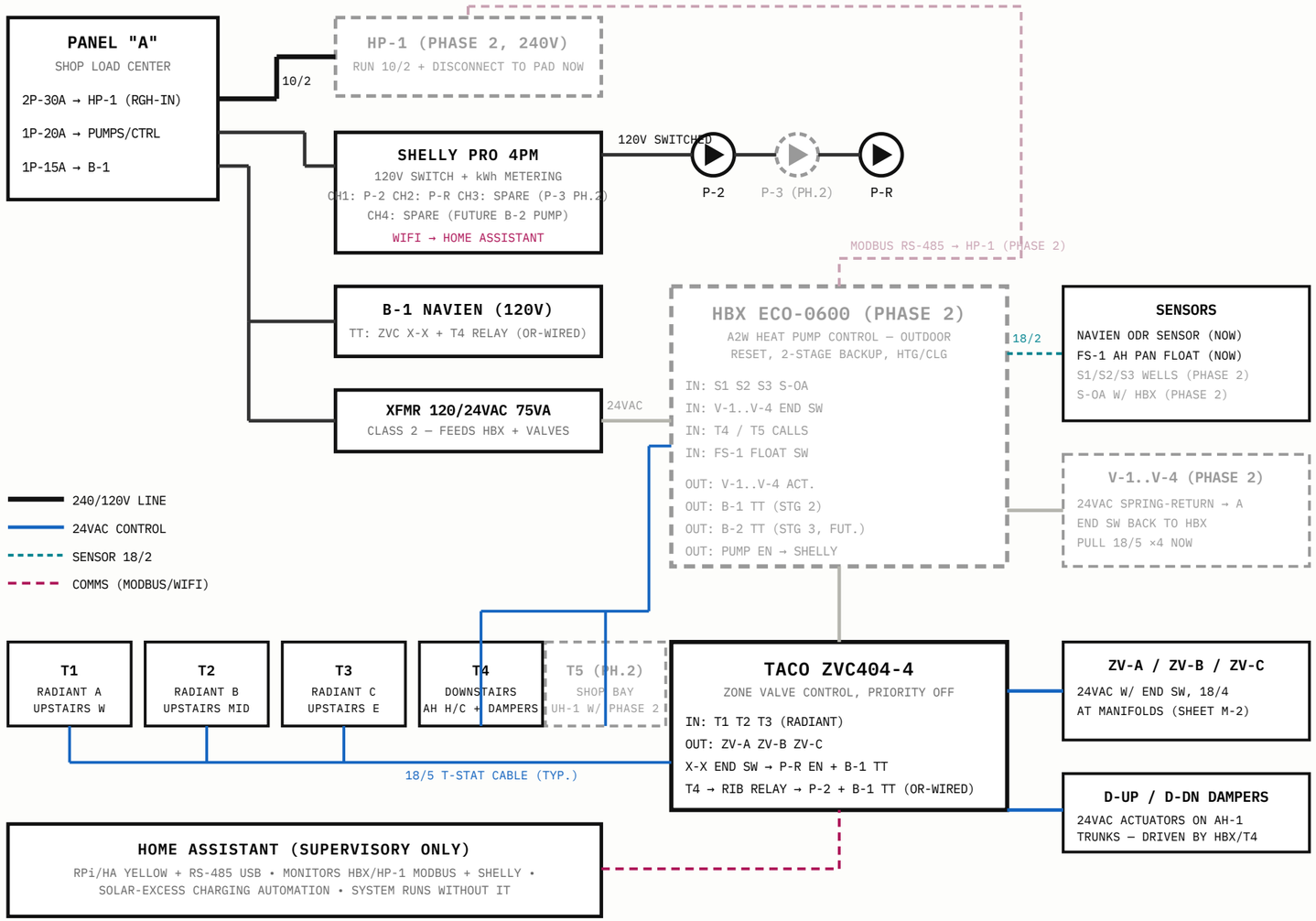
- PHASE 1: NAVIEN ODR + ZVC404 + RELAYS — NO CENTRAL CONTROLLER NEEDED.
- SENSORS NOW: NAVIEN OUTDOOR SENSOR (N WALL) + FS-1 FLOAT. S1/S2/S3 WELLS ROUGHED IN.
- PHASE 2: ADD HBX ECO-0600 + V-1..V-4 + HP-1 MODBUS — CONDUIT + SPARE 18/5 PULLED NOW.
- PUMPS P-2/P-R: 120V VIA SHELLY/RELAYS; B-1 TT DRY CONTACT. P-3 ARRIVES WITH UH-1.
- PRIORITY: DHW (COMBI INTERNAL) > SPACE HEAT. BURST/COOLING PRIORITIES ARRIVE IN PHASE 2.
- FS-1 HARDWIRED TO BREAK FUTURE AH COOLING CALL. RADIANT HARD INTERLOCK INSTALLED WITH PHASE 2 VALVES.

FUTURE — SOLAR-DRIVEN CHARGING

- WHEN PV EXPORT > HP-1 DRAW: HA FORCES HP-1 TO CHARGE BT-1 TO 120°F CAP (HTG) OR PRE-CHILL TO 42°F (CLG). CONTROL-ONLY — NO EXTRA PIPING.
- TANK = THERMAL BATTERY: 250 GAL × 20°F ≈ 41,000 BTU/CYCLE.
- FUTURE B-2 STUBS (SHEET M-3) VALVED + CAPPED FOR SECOND BOILER.

CONTROLS — WIRING & THERMOSTATS

LINE VOLTAGE + 24VAC CONTROL + COMMS • SEPARATE CONDUITS FOR LINE vs LOW VOLTAGE • NTS



CONTROL HARDWARE SCHEDULE

HBX ECO-0600 A2W CONTROL — PHASE 2, W/ HP-1	(1)
TACO ZVC404-4 ZONE VALVE CONTROL	1
SHELLY PRO 4PM (PUMP SWITCHING + METERING)	1
24V H/C PROGRAMMABLE T-STATS (T1–T4 NOW, T5 PH.2)	4 + (1)
NAVIEN ODR SENSOR (INCL.) + 10K WELLS (PH.2)	1 + (3)
24VAC DAMPER ACTUATORS (D-UP/D-DN)	2
AH PAN FLOAT SWITCH (FS-1)	1
75VA CLASS-2 TRANSFORMER + RIB RELAYS	1 + 2
HA YELLOW / RPi + RS-485 ADAPTER (OPTIONAL)	1

THERMOSTAT SCHEDULE

ID	LOCATION	CONTROLS
T1	UPSTAIRS WEST	ZV-A (RADIANT ZONE A)
T2	UPSTAIRS MID	ZV-B (RADIANT ZONE B)
T3	UPSTAIRS EAST	ZV-C (RADIANT ZONE C)
T4	DOWNSTAIRS	AH-1 HTG + D-UP/D-DN (CLG IN PH.2)
T5	SHOP BAY	FUTURE — UH-1 WITH PHASE 2

WIRING NOTES

1. LINE (120/240V) AND CLASS-2 (24V/COMMS) IN SEPARATE CONDUITS; 18AWG MIN CONTROL WIRE.
2. PULL SPARE 18/5 + RS-485 TO HP PAD, TANK SPACE, UH LOCATION NOW — WALLS OPEN.
3. FS-1 BREAKS FUTURE AH COOLING CALL DIRECTLY (HARDWIRE, NOT SOFTWARE).
4. B-1 TT DRIVEN BY ZVC X-X AND T4 RELAY, OR-WIRED THROUGH ISOLATION RELAYS.
5. PHASE 2: HBX TAKES OVER STAGING + RADIANT HARD LOCKOUT; HP-1 ON MODBUS.
6. LABEL EVERY CONDUCTOR AT BOTH ENDS PER THIS SHEET'S TAGS.

FUTURE B-2 — SECOND BOILER (IF B-1 CAN'T KEEP UP)

- TRIGGER TO ADD: B-1 DESIGN-DAY DUTY > 90% OR BURST RECOVERY > 45 MIN (HA LOGS BURN HOURS VIA SHELLY/MODBUS).
- UNIT: NAVIEN NFB-200H HTG-ONLY (DHW STAYS ON B-1). PIPE PARALLEL TO B-1 AT THE VALVED + CAPPED TEES (SHEET M-3), OWN CHECK + UNIONS + CIRC; UPSIZE BOILER GROUP TO 1-1/4".
- CONTROL: HBX STAGE-3 DRY CONTACT (TERMINAL RESERVED, WIRED TO J-BOX AT STUBS). SHELLY CH4 = B-2 PUMP. LEAD/LAG ROTATION VIA HA.
- ROUGH-IN NOW: 18/2 TO STUB J-BOX, CONDUIT SPACE, VENT + GAS TEE CAPPED.